Assignment: Binary Image Classification of Jackfruit Leaf and Mango Leaf

You are required to design, implement, train, and evaluate a **binary image classifier** that distinguishes between **jackfruit leaves** and **mango leaves**.

Write a report including answers to the following questions:

1. Draw the architecture of a **Multi-Layer Perceptron (MLP)** with **three hidden layers** containing **8, 6, and 4 neurons**, respectively, for classifying images of **jackfruit leaves** and **mango leaves**. Clearly show the input layer, hidden layers, and output layer.
2. Calculate the **total number of trainable parameters** (weights and biases) in your designed MLP.
3. Draw the architecture of a **Convolutional Neural Network (CNN)** with **three convolutional hidden layers**. Each convolutional layer should use **eight 3 × 3 filters**. Clearly show the number and size of the feature maps at each layer. State any assumptions you make regarding the padding and stride.
4. Write a **Python program** to implement the CNN architecture designed in Question 3. Your code should include:
    - Model definition
    - Model compilation
    - Model summary
5. Suppose you captured **5,000 labeled leaf images using your smart phones**. Determine how many images you would allocate to the **training**, **validation**, and **test** sets. Justify your chosen data split.
6. What **loss function** would you use to train this binary classifier? Justify your choice.
7. Which **activation function** would you use in:
    - The hidden layers?
    - The output layer?
8. Justify your choices.
9. Write the format of the target labels (**y**) for this binary classification problem. Show how the labels for **jackfruit leaf** and **mango leaf** would be represented.
10. How can you reduce the number of trainable parameters in your MLP? Discuss at least **three approaches**, and explain the advantages and disadvantages of each.
11. Which **evaluation metrics** would you use to assess the performance of your classifier? Explain why each metric is appropriate for binary classification.
12. Draw and explain a **confusion matrix** for this binary classification problem. Define **True Positive (TP), True Negative (TN), False Positive (FP),** and **False Negative (FN)** in the context of classifying jackfruit and mango leaves.
13. Explain the difference between **binary cross-entropy** and **categorical cross-entropy**. Which one is more appropriate for this problem, and why?

14. If your classifier achieves **99% training accuracy** but only **84% test accuracy**, what problem is likely occurring? Discuss at least **four techniques** that can help address this issue.
15. Compare the advantages and disadvantages of using an **MLP** and a **CNN** for leaf image classification.
16. Suppose your CNN incorrectly classifies several mango leaves as jackfruit leaves. Suggest possible reasons for these misclassifications and propose appropriate solutions.
17. Propose **three improvements** that could increase the accuracy and robustness of your binary classifier.